

# Pump-beam-instability limits to Raman-gain-doublet “fast-light” pulse propagation

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We investigate the behavior of a system for generating “fast-light” pulses in which a bichromatic Raman pumping beam is used to generate optical gain at two frequencies and a region of anomalous dispersion between them. It is expected that increasing the gain will increase the pulse advancement. However, as the gain increases, the pumping field becomes increasingly distorted, effectively limiting the pulse advancement. We observe as much as 12% of the input pump power converted to orthogonal polarization, broadening of the initially bichromatic pump field (25 MHz initial frequency separation) to more than 2.5 GHz, and a temporal collapse of the pump beam into an erratic train of sub-500-ps pulses. The instability is attributed to the combined effects of the cross modulation instability and stimulated Raman scattering. Extreme distortion of an injected pulse that should (absent the instability) experience an advancement of 21% of its width is observed. We conclude that the fast-light pulse advancement is limited to just a few percent of the pulse width using this pulse advancement technique. The limitation imposed by the instability is important because careful study of the information velocity in fast-light pulses requires that pulse advancement be large enough to distinguish the velocities of different pulse features. Possible methods for achieving pulse advancement by avoiding the distortion caused by the instability are discussed.

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## I. INTRODUCTION

Using electromagnetic fields to create large atomic coherences, it is now possible to tailor the absorption, amplification, and dispersion properties of multilevel atoms [1,2]. One application of these techniques is to adjust the group velocity  $v_g$  of light pulses to make them propagate very slowly ( $v_g \ll c$ ) [3–6] or very fast ( $v_g > c$  or  $v_g < 0$ ) [4,7–10]. A recent technique for creating “fast-light” pulses uses a continuous-wave bichromatic Raman pump laser beam to tailor the dispersion properties of an atomic vapor [8,9,11], resulting in two gain features with slightly different frequencies. Between this gain doublet, the phase-velocity dispersion is anomalous and the group-velocity dispersion can be minimized [9,11,12]. As a result, a pulse traveling through the medium experiences so-called “fast-light” propagation with little distortion if the pulse spectral bandwidth falls between the doublet [2]. Note that the pulse experiences optical amplification rather than absorption as it would if it were passing through the anomalous-dispersion spectral region characteristic of a resonant absorber [4,7,10,13–16].

Fast-light pulse propagation has been the subject of extensive theoretical investigation for nearly a century because of its potential implications for the special theory of relativity. Early work by Brillouin and Sommerfeld [17] considered the propagation of a pulse with a rectangular pulse envelope whose frequency was tuned to the center of an absorption resonance where  $v_g > c$  or  $v_g < 0$ . It was found that the front (i.e., the leading discontinuity in the pulse amplitude) travels at  $c$  and the pulse envelope becomes distorted. One possible interpretation of this finding, appearing in some textbooks, is that the group velocity loses its meaning for fast-light media.

In the 1970s, Garrett and McCumber [18] found that Gaussian-shaped pulses can propagate through an absorber with little distortion when  $v_g > c$  or  $v_g < 0$ . The amount that the peak of the pulse is advanced at the exit of the medium of length  $L$  in comparison to a pulse propagating through vacuum over the same distance is given by

$$t_{adv} = \frac{L}{c} - \frac{L}{v_g}, \quad (1)$$

where  $v_g = 1/k_1$ ,

$$k_m = \left. \frac{d^m k(\omega)}{d\omega^m} \right|_{\omega=\omega_c}, \quad (2)$$

$k(\omega)$  is the magnitude of the frequency-dependent wave vector, and  $\omega_c$  is the carrier frequency of the pulse. Such fast-light pulse advancement occurs due to the rephasing of the frequencies of the wave packet due to the causal response of the atoms to the applied field and hence should be entirely consistent with the special theory of relativity. Garret and McCumber’s prediction has been confirmed experimentally for pulse propagation through an absorber [4,7,10,13–16], a medium with a gain doublet [8,9], in electrical pulse propagation along a dispersion-managed cable [19], and through low-frequency electronic filters [20], although the interpretation of the results has sometimes been controversial [21].

Detailed theoretical analysis of fast-light pulse propagation by Chiao and Steinberg [22] has revealed that it is possible to make essentially all velocities associated with propagating pulses (group, energy, signal, etc.) greater than  $c$  with little pulse distortion [23]. On the other hand, they have found that a point of nonanalyticity (e.g., a discontinuity) in the envelope of a pulse always propagates at  $c$ , consistent

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with the earlier findings of Brillouin and Sommerfeld [17]. Hence, they propose that such a point is associated with “new information” carried by the pulse to ensure that pulse propagation in fast-light media is consistent with the special theory of relativity. For pulses containing only a few photons, it is predicted that the point of constant signal-to-noise ratio also travels at  $c$  or slower [24–26], indicating that quantum fluctuations might limit the information velocity to speeds equal to or less than  $c$  even when the pulse is smooth. We note that there is some controversy as to whether a true point of nonanalyticity can be created in the laboratory because of the finite bandwidth of pulse shapers and switches [27]. We also note that the issue of the speed at which information travels between two spatial locations has not yet been addressed by the information theory community to the best of our knowledge [28].

To most clearly test these predictions in an experiment, it is necessary to have large relative pulse advancement  $\mathcal{A} \equiv t_{adv}/t_p$ , where  $t_p$  is the pulse width [full width at half maximum (FWHM)], so that the propagation of points of nonanalyticity can be distinguished from other pulse features. For the gain-doublet medium suggested by Steinberg and Chiao [11] and studied experimentally [8,9], the fast-light pulse advancement is proportional to the gain, as shown in the next section. As the gain (and hence  $\mathcal{A}$ ) increases, it has been suggested that the quantum-initiated process of superfluorescence or amplified spontaneous emission will set in, thus obscuring or limiting the pulse advancement [24–26]. Increasing the gain also requires reduction of the input pulse intensity so that the gain is not saturated [29].

The primary purpose of this paper is to demonstrate that the modulation instability, induced by the intense bichromatic field used to dispersion-tailor the atoms, imposes a severe limitation to the achievable pulse advancement  $\mathcal{A}$  and sets in well before the processes of superfluorescence and amplified spontaneous emission [25]. When the gain is high, the instability causes the bichromatic field to undergo a fantastic spectral broadening and temporal collapse as it propagates through the vapor cell. As a result, the pump field evolves from a purely bichromatic field at the entrance face of the medium to a very broadband multifrequency field at the exit, so that the atom-field dispersion properties experienced by the pulse vary spatially and are very different from the ideal picture described above.

In the next section, we show that gain-assisted fast-light pulse advancement is proportional to the gain and give an estimate of the expected size of  $\mathcal{A}$  for reasonable choices for the experimental parameters. Section III describes our experimental setup, while Sec. IV details our observation of the spectral broadening and temporal collapse of the bichromatic pump field as it propagates through a potassium vapor and our interpretation of this effect in terms of what is known as the cross (or induced) modulation instability. We then demonstrate in Sec. V how the modulation instability causes severe distortion of pulses propagating through the driven vapor, thereby limiting the achievable fast-light pulse advancement. Finally, we suggest in Sec. VI methods for suppressing the instability.

## II. GAIN-ASSISTED “FAST-LIGHT” PULSE ADVANCEMENT

To see why gain-assisted fast-light pulse advancement is proportional to the gain, consider two Lorentzian gain lines, each with a line-center intensity gain coefficient  $g_0$ , half width at half maximum (HWHM)  $\gamma$ , and center frequency  $\omega_{\pm} = \omega_a \pm \delta/2$ . The magnitude of the frequency-dependent wave vector for the medium is given by

$$k(\omega) = \frac{\omega}{c} n_h + \frac{g_0 \gamma}{2} \left[ \frac{1}{\omega - \omega_- - i\gamma} + \frac{1}{\omega - \omega_+ - i\gamma} \right], \quad (3)$$

where  $n_h$  is the background refractive index of the host material. In an atomic vapor where  $n_h = 1$ , the group-velocity dispersion term  $k_3$  can be eliminated by choosing

$$\delta = 2(1 + \sqrt{2})\gamma, \quad (4)$$

leading to

$$t_{adv} = \frac{g_0 L}{\gamma} \frac{\sqrt{2} - 1}{4}. \quad (5)$$

Note that the second-order dispersive term  $k_2$  cannot be eliminated by adjusting the atomic parameters; it is purely imaginary and gives rise to pulse compression [9,12,15]. The relative pulse advancement  $\mathcal{A}$  can be determined from Eq. (5) and a knowledge of the pulse duration (and hence the spectral bandwidth). It is not possible to increase  $\mathcal{A}$  just by decreasing  $t_p$  because, for very short pulses, some frequencies will fall outside the region of large anomalous dispersion, giving rise to pulse distortion. The extent of the distortion depends on the atomic parameters, the specific pulse shape and bandwidth, and the gain path length  $g_0 L$ . For a Gaussian-shaped pulse, a reasonable compromise between distortion and advancement is to set the width (FWHM) of the pulse power spectrum to  $\delta/6$  so that most of the pulse spectral power falls within the gain doublet. Under this condition, we find that the distortion is less than 5% of the peak height up to a line-center gain path length of  $g_0 L = 12$ .

Given this choice for the pulse shape and spectral width, the relative pulse advancement is given approximately as

$$\mathcal{A} = \frac{t_{adv}}{t_p} \approx 0.03 g_0 L. \quad (6)$$

Equation (6) shows that the relative advancement is simply proportional to the gain path length. We emphasize that this result is intended only to demonstrate the fundamental link between pulse advancement and gain in this system. Modest improvement may be possible with slightly different choices for  $\delta/\gamma$ , pulse shape, etc., but significantly more advancement can be achieved only by changing the technique or relaxing the distortion constraints [9,12].

To realize large relative pulse advancement with little pulse distortion, it is important to use a system where the doublet splitting  $\delta$  can be adjusted to satisfy condition (4). As pointed out by Steinberg and Chiao [11], one method for obtaining a gain doublet is to use the atomic Raman scatter-

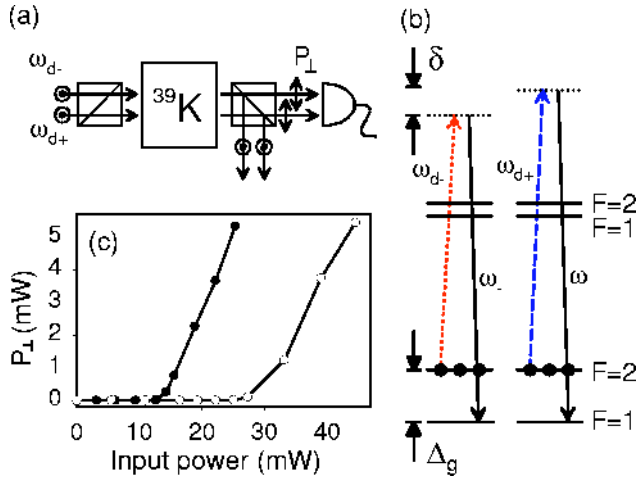


FIG. 1. (a) Experimental setup. (b) Level diagram showing the Raman anti-Stokes processes occurring for both pump frequencies. The population is pumped into the upper ( $F=2$ ) ground state by the same bichromatic pumping beam. (c) Total generated power in the orthogonal polarization as a function of input power in each frequency. Closed (open) circles represent data for a bichromatic (monochromatic) beam.

ing process occurring in optically pumped and laser driven alkali-metal atoms. In this process, Raman pump photons are annihilated and probe-pulse photons created as the atom makes a transition between ground-state Zeeman hyperfine levels that are inverted due to an auxiliary optical pumping process. A gain doublet is produced when the Raman pumping beam is bichromatic, where the doublet spacing is given by the frequency difference of the bichromatic field and the resonance width is set by the ground-state coherence decay rate. In addition, it has been shown by Concannon *et al.* [30] that extremely large laser beam amplification can be realized using this technique.

Steinberg and Chiao's proposal was first implemented experimentally by Wang *et al.* [8], who used an optically pumped and bichromatically driven rubidium vapor. Because they used a hot vapor, the Raman pump beam copropagated with the probe pulse to cancel Doppler broadening of the gain resonances. For the available laser power, they were able to obtain  $g_0L \approx 1.35$  and  $\mathcal{A} \approx 2.6\%$ . [This differs slightly from the  $\mathcal{A} \approx 4\%$  predicted by Eq. (6) due to their different choices for  $\delta$  and  $t_p$ .] Because  $\mathcal{A}$  is so small in their experiment, it would be difficult to test predictions concerning the information propagation velocity using their setup.

### III. HIGH-GAIN ATOMIC RAMAN SCATTERING FOR FAST-LIGHT APPLICATIONS

In an attempt to achieve large relative pulse advancement, we have used a different experimental apparatus [Fig. 1(a)] that is capable of producing large gain via the atomic Raman scattering process. We use potassium atoms in our setup because the small hyperfine ground-state splitting ( $\Delta_g = 462$  MHz) yields large Raman gain and we use a linearly polarized drive laser beam that simultaneously optically pumps the atoms into one hyperfine level and drives the

Raman scattering process [30]. With this setup, described in greater detail below, we can routinely obtain  $g_0L$  in excess of 15, which should give  $\mathcal{A} \approx 0.45$  according to Eq. (6).

Unfortunately, we find that the competing nonlinear optical process known as the modulation instability occurs as the Raman gain is increased. The instability is enhanced when the pump field is bichromatic, and is then often referred to as the cross (or induced) modulation instability (CMI) [31]. In the present context, where large atomic coherence is created by the electromagnetic field, the CMI leads to extreme frequency broadening of the Raman pump beams even for low pump beam powers.

To observe and characterize the limitations imposed by this competing nonlinear optical effect, we pass a bichromatic field through a potassium vapor, which is contained in a 20-cm-long glass cell heated to  $130^\circ\text{C}$ , corresponding to an atomic number density of  $4 \times 10^{12} \text{ cm}^{-3}$ . The cell has no paraffin coating on the interior walls to prevent depolarization of the ground-state coherence, nor does it contain a buffer gas that would slow diffusion of atoms out of the pump laser beam. The cell reflects 12% of the incident light at each window.

The atoms are pumped by two laser beams tuned to  $\omega_{d\pm} = \omega_{11} + 2\pi(1.67 \text{ GHz}) \pm \delta/2$ , where  $\omega_{11}$  is the  $^{39}\text{K } 4^2S_{1/2} (F=1)$  to  $4^2P_{1/2} (F=1)$  transition frequency, as shown in Fig. 1(b). The pump frequencies are chosen to balance favorably the stimulated Raman scattering, optical pumping, and absorption at the operating temperature. The beams are generated using a continuous-wave Ti:sapphire ring laser (Coherent 899-21) and two acousto-optic modulators (AOMs). They are recombined and pass through a single-mode optical fiber to remove spatial distortion introduced by the AOMs and to ensure that they have the same spatial mode. We place a polarizing beam splitter (PBS) after the fiber and collimate the beam at  $450 \mu\text{m}$  ( $1/e$  field radius).

The pump beams perform a dual role in the experiment. They optically pump the atoms into the upper ( $F=2$ ) ground state and provide the drive field necessary for stimulated Raman scattering (Raman gain) [30]. Maximum gain occurs for probe frequencies of  $\omega_{\pm} = \omega_{d\pm} + \Delta_g$ , as shown in Fig. 1(b). Selection rules dictate that light with orthogonal linear polarization will experience gain [32] for a linearly polarized Raman pump.

The resonance half width (HWHM)  $\gamma$  of the Raman gain features is weakly intensity dependent for the powers used in our experiment and is approximately equal to  $2\pi(2.3 \text{ MHz})$ . We set  $\delta/2\pi = 25 \text{ MHz}$ , which is somewhat larger than that required by Eq. (4) so that we can more easily resolve this frequency difference with our optical spectrum analyzer (see below). Because the nature of the modulation instability is found to be insensitive to the choice of  $\delta$  when it is in the range of tens of MHz, such an adjustment does not affect our conclusions.

A continuous-wave or pulsed probe field (carrier frequency  $\omega_c$ ) can also be injected into the cell with polarization orthogonal to the pump polarization to measure the Raman gain or to measure pulse propagation in the vicinity of the gain doublet. The probe beam originates from the same

laser, travels through a double-pass AOM and a single-mode fiber, and is combined with the pump beams via the PBS.

The light emerging from the vapor cell passes through a rotatable linear polarizer and can be sent to a confocal optical spectrum analyzer (Coherent 240, 7.5 GHz free spectral range, 15 MHz resolution) or fast detector (New Focus 1537, 6 GHz bandwidth). The voltage produced by the detector can then be observed with either a fast oscilloscope (Tektronix TDS 680B, 1 GHz analog bandwidth,  $5 \times 10^9$  samples/s sample rate) or an electronic spectrum analyzer (HP 8566B).

#### IV. CHARACTERISTICS OF THE LIGHT GENERATED BY THE MODULATION INSTABILITY

When the bichromatic pump beam is applied to the cell with no probe beam present, the pump light is modified dramatically by nonlinear interactions with the atoms. Figure 1(c) shows the power that emerges from the cell in the orthogonal linear polarization (relative to the input pump polarization) as a function of the input pump power in each beam. For low pump power, there is little power converted to the orthogonal polarization, as expected for the case when the effects of the instability are negligible. However, as the pump power increases, an increasing fraction of the total input light is converted to the orthogonal polarization due to the instability. We call this fraction the conversion efficiency. This polarization change is due to the CMI and Raman processes, which can create photons with the polarization orthogonal to the pump polarization [32]. The threshold for this effect (arbitrarily defined as the input power which yields 0.1% conversion efficiency) occurs at approximately 11 mW per frequency component (22 mW total pump power), which corresponds to a probe beam intensity gain of  $e^{12.5}$ , measured by injecting a weak probe at  $\omega_{d\pm} + \Delta_g$ . Increasing the pump power by a factor of 2 (gain greater than  $e^{15}$ ) increases the conversion efficiency to 12%.

For comparison, the threshold for a monochromatic Raman pump beam occurs at 25 mW, which is considerably higher than even the *total* input power for the bichromatic threshold [Fig. 1(c)]. This confirms that the bichromatic nature of the light is an essential part of the effect.

Returning to the bichromatic pump case, we find that the light emerging from the vapor cell in the orthogonal polarization has dramatically modified frequency properties. Figure 2(a) shows the optical spectrum of the orthogonally polarized generated light at the maximum input power of 22 mW per frequency component, as described above. It is seen that most of the generated light occurs near  $\omega_{d\pm} + \Delta_g$ , the spectral region of maximum Raman gain [32]. There are also notable frequency components near  $\omega_{d\pm} - \Delta_g$  and  $\omega_{d\pm} + 3\Delta_g$ . Note that the peaks are extremely broad compared to the input light and have a series of spikes spaced at the pump beam frequency separation  $\delta$ .

Our interpretation of the observed spectral broadening is that it is due to a combination of Raman and CMI effects. The locations of the broad peaks can be explained by the Raman scattering processes shown diagrammatically in Fig. 2(b). The large peak at  $\omega_{d\pm} + \Delta_g$  is partly due to the Raman anti-Stokes effect, and the peak at  $\omega_{d\pm} - \Delta_g$  is partly due to

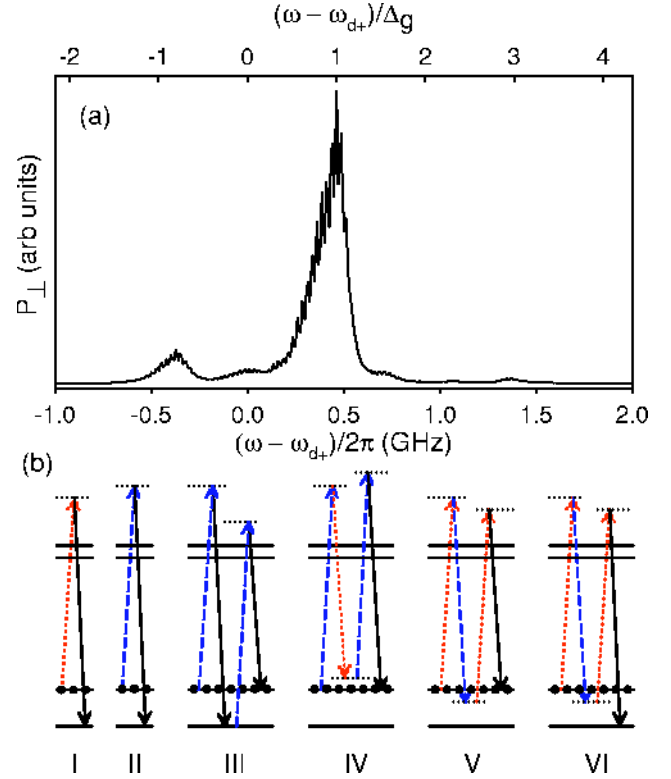


FIG. 2. (a) Optical spectrum of the generated light in the orthogonal polarization with 22 mW in each input frequency. The top scale shows the frequency in units of the ground-state splitting. (b) Level diagrams showing Raman anti-Stokes scattering (I and II), Raman Stokes/anti-Stokes coupling (III), the CMI (IV and V), and a mixed Raman and CMI effect (VI). In each case, the dotted (dashed) arrows correspond to  $\nu_-$  ( $\nu_+$ ) photons, and the solid arrows correspond to generated photons.

the Raman Stokes/anti-Stokes coupling [33]. Note that the blueshifted anti-Stokes line is much stronger than the corresponding Stokes line because the optical pumping process causes the upper ( $F=2$ ) ground state to be more highly populated than the lower ( $F=1$ ) ground state. This ground-state inversion is different from the effect in common thermalized systems that experience gain at the Stokes line, rather than the anti-Stokes line.

The smaller peak at  $\omega_{d\pm} + 3\Delta_g$  is a result of repeated Raman scattering. Light generated by Raman scattering can in turn drive further Raman scattering. Every time a Raman scattering event occurs, a photon is effectively shifted in frequency by  $\Delta_g$ , but it is also shifted in polarization by  $90^\circ$ , as described above [32]. Because our apparatus includes a polarizer between the cell and detector, only odd-order peaks (those that have been Raman shifted an odd number of times) appear. When we rotate the polarizer by  $90^\circ$  so that it is aligned with the input pump light (data not shown), we observe that only even orders pass through the polarizer.

Raman processes alone do not completely explain our observations because Raman processes normally produce extremely narrow lines [32], whereas the lines shown in Fig. 2(a) are very broad. Our interpretation is that the width and shape of these broad peaks are a result of the CMI. The



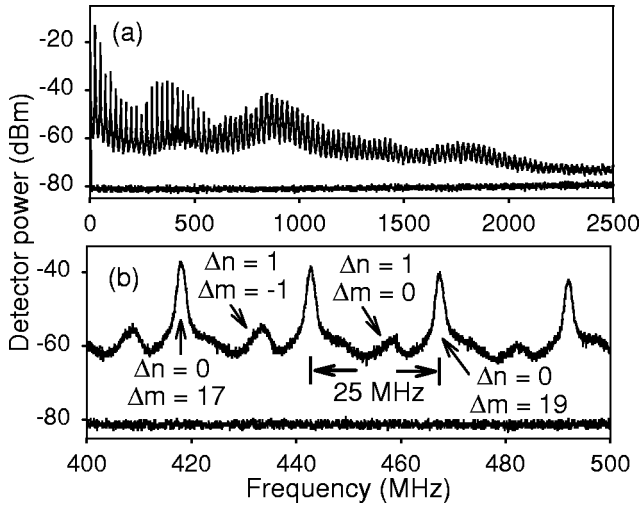


FIG. 3. (a) Low- and (b) high-resolution electronic power spectra of the voltage generated by the fast detector that senses the light generated in the orthogonal polarization.

four-wave mixing process that causes the CMI [shown in Fig. 2(b)] is an efficient mechanism for translating existing frequencies by the pump separation  $\delta$ . The process involves two photons of one pump frequency, one photon of the other pump frequency, and a newly generated photon. This process can generate photons that are either shifted up by an amount  $\omega_{d+} - \omega_{d-} + \omega_{d+} = \omega_{d+} + \delta$  [IV of Fig. 2(b)] or down by an amount  $\omega_{d-} - \omega_{d+} + \omega_{d-} = \omega_{d-} - \delta$  [V of Fig. 2(b)] [31]. Like the Raman processes, the CMI can also occur repeatedly, creating multiple sidebands. For example, a generated photon of frequency  $\omega_{d+} + \delta$  can interact with two  $\omega_{d-}$  photons to produce a photon of frequency  $\omega_{d-} - (\omega_{d+} + \delta) + \omega_{d-} = \omega_{d-} - 2\delta$ . There are also processes that incorporate elements of both the Raman and CMI processes, which produce shifts of  $\pm\delta \pm \Delta_g$  [VI of Fig. 2(b)]. Because these processes occur repeatedly and shift frequencies in both directions, the otherwise narrow Raman lines become dramatically broadened into a comb with frequencies  $\omega_{d\pm} + n\Delta_g + m\delta$  for integer  $n$  and  $m$  [34,35]. Due to the resonant nature of the interaction, the power in each frequency must diminish with increasing  $|n|$  and  $|m|$ , as observed in Fig. 2(a). This combined CMI and Raman effect is very similar to one that has been exploited by, for example, Radic *et al.* [36] in optical fiber amplifiers to achieve broad flat gain over enormous (22 nm) bandwidths.

To achieve greater frequency resolution, we focus the generated light on the fast detector and observe the detector output with the electronic spectrum analyzer, which reveals frequency *differences* that are present in the optical field. As seen in Fig. 3(a), the spectrum of the generated photocurrent extends to several gigahertz and contains broad features at multiples of  $\Delta_g$  resulting from the Raman effect. As in the optical spectrum analyzer results [Fig. 2(a)], there are peaks spaced at the pump splitting ( $\delta = 25$  MHz) that are due to the CMI. This is more clearly seen in Fig. 3(b), which shows a smaller frequency range and is recorded with a resolution bandwidth of 1 MHz. Here, the peaks have been labeled according to the optical frequencies that they correspond to.

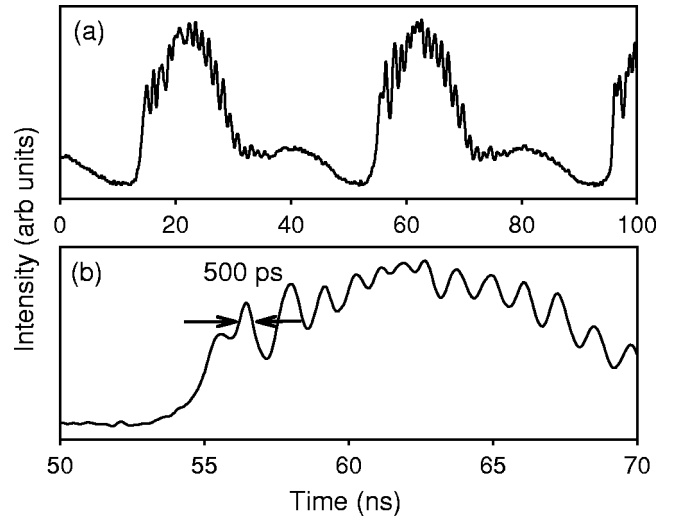


FIG. 4. Temporal evolution of the light generated in the orthogonal polarization at (a) slow and (b) fast time scales.

Note that the  $\Delta n=1, \Delta m=0$  peak occurs at less than the ground-state splitting 462 MHz. We attribute this to ac Stark shifts of the levels. The  $\Delta n=1$  peaks are smaller because they are the results of interference between optical components with mutually orthogonal polarizations, one of which is suppressed by the polarizer.

Figure 3(b) also shows that the power *between* the peaks does not fall to the electronic noise floor, as would be expected for a discrete comb of frequencies. Instead, there are no quiet frequency windows over a range of more than 2 GHz. The power remains 20 dB above the noise floor and more than 30 dB above the standard quantum limit in the range from dc to  $\sim 1$  GHz. We observe that variation of  $\delta$  has no significant effect on the spectrum, which suggests that the presence of this interpeak power is not simply related to the commensurability of  $\delta$  and  $\Delta_g$ . Harris and co-workers have studied a very similar system in deuterium in which they set  $\delta = \Delta_g$  and exploit the CMI and Raman generation processes to create very short pulses [34,35]. Their theory predicts a frequency comb, but does not predict the continuous spectrum we observe. This interpeak power may be related to effects observed by Lukin based on the creation of fast-modulating electromagnetically induced transparency windows for the driving field [37]. We note that a broad continuous spectrum often indicates chaotic temporal evolution. Chaotic behavior seems quite plausible given that much of the generated light results from instabilities in a highly nonlinear system.

To observe the temporal evolution of the generated light, we record the output of the detector directly with a fast oscilloscope as shown in Fig. 4, where it is seen that the intensity evolves in a complex pattern that roughly repeats with a period of 40 ns. While this corresponds to a frequency of 25 MHz, it should be noted that this is not simply beating between the two pumping beams (which are not transmitted through the polarizer) but is a result of interference between frequency components of the generated field. There is also higher-speed structure [Fig. 4(b)] that includes pulses as

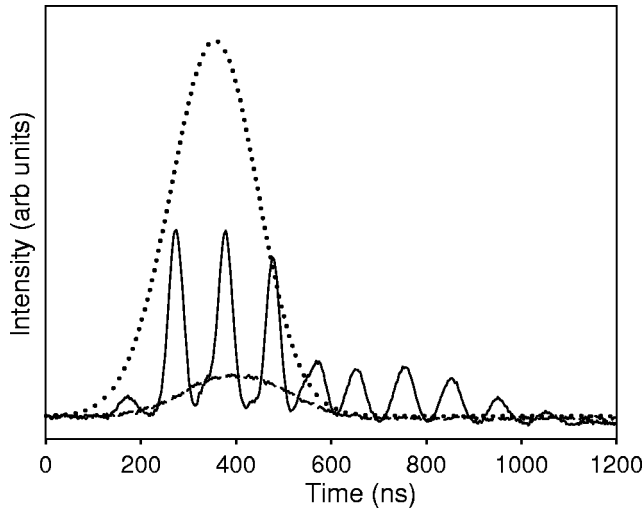


FIG. 5. Pulse propagation through the fast-light medium experiencing CMI-based pump distortion (solid line). The pulse is both stretched and modulated at the pump separation frequency 10 MHz. An identical pulse propagating through vacuum (dashed line) and a pulse as predicted by an exact numerical calculation (dotted line) are shown for comparison.

short as 500 ps, which is at the analog-bandwidth limit of our oscilloscope.

To determine whether the observed fluctuations are a manifestation of deterministic chaos, we used the false nearest neighbor and global Lyapunov exponent nonlinear statistical analysis methods [38]. Unfortunately, these tests failed to converge to a meaningful result due to the limited vertical resolution of our oscilloscope ( $\leq 8$  bits).

We emphasize that our observations depend strongly on the presence of a bichromatic pumping beam and are dramatically different when a monochromatic pump beam is used. With only a single frequency, the Raman process still occurs, but the CMI does not. As a result, the optical spectrum consists of narrow peaks (similar to the input beam in width) spaced by the ground-state splitting  $\Delta_g$ . As expected, the electronic spectrum of the fast-detector signal consists of strong peaks at integer multiples of  $\Delta_g$ , with the power quickly dropping to the noise floor between the peaks.

## V. FAST-LIGHT LIMITS DUE TO THE MODULATION INSTABILITY

To demonstrate the significance of these effects on pulse propagation, we inject a pulse into a potassium vapor in which a bichromatic Raman pump creates a gain doublet, as described above. We set  $\delta = 10$  MHz to satisfy condition (4) and use a 246 ns pulse so that the bandwidth is equal to  $\delta/5.57$ .

To avoid as much as possible the deleterious effects of the modulation instability, we set  $g_0 L \approx 7.0$ , which is far below the threshold value for the modulation instability described in Sec. IV. At this gain path length, we expect  $\mathcal{A} \approx 21\%$  and that the pulse distortion due to linear dispersion should be negligible.

Figure 5 shows the experimentally measured Gaussian-envelope pulse after propagating through vacuum (dashed line) and the theoretically predicted output pulse based on the theory outlined in Sec. II (dotted line). The theoretical pulse is calculated numerically and is exact to all orders in  $k(\omega)$ , but does not account for nonlinear effects such as the CMI. The solid line shows the experimentally observed output pulse, which is distorted dramatically. The pulse envelope is modulated at 10 MHz, which is equal to the frequency separation of the bichromatic pump beam. The fact that the observed output pulse is very different from that expected by the linear dispersion theory indicates that the CMI is responsible for the distortion, even though we have lowered the pump beam power to below the nominal threshold for the instability. The instability remains a factor because it is now seeded by probe photons rather than quantum fluctuations, effectively lowering the instability threshold.

Similar pulse-propagation behavior is observed for lower Raman gain, becoming less pronounced only when the gain is so low that the relative advancement is as small as that observed by Wang *et al.* [8].

## VI. DISCUSSION

Our experiments demonstrate that the modulation instability places severe restrictions on the fast-light process. These restrictions are of a fundamentally different nature than those due to linear dispersive effects described previously [9,12]. While the CMI does not directly limit the group velocity, it does restrict the practically achievable pulse advancement to just a few percent of the pulse duration.

There may be several techniques for reducing or altogether avoiding pulse distortion from the CMI. Considerable success has been achieved using the anomalous dispersion of an absorption line [13,16] and due to electromagnetically induced absorption [7,10], although the advanced pulses experience significant absorption that might be unacceptable for some applications.

In the gain-assisted system considered here, the CMI could be suppressed using an orthogonal pump-probe geometry if the Doppler effect is canceled. Doppler broadening can be reduced using an atomic beam source of atoms, or with a gas of cold trapped atoms such as that used in some of the “slow-light” experiments of Hau *et al.* [3].

Yet another possible technique for suppressing the CMI is to use two separate spatial regions for the two Raman pump beams, where the probe pulse passes through each region in series [39]. This method retains the favorable advantages of the bichromatic Raman technique (large gain and adjustable resonance frequencies), while suppressing the CMI by ensuring that both beams never interact simultaneously with the medium. We are currently investigating this method.

While it is difficult to predict the behavior of such complex nonlinear systems, we expect this fast-light pulse propagation technique with the CMI suppressed will be limited ultimately by processes such as amplified spontaneous emission and superfluorescence at values of  $g_0 L \approx 30$ . At such

high gain, single-photon quantum fluctuations are amplified to macroscopic intensities and can obscure the advanced pulse [25,26], saturate the transition [29], and deplete the pumping beams. Inserting  $g_0 L \approx 30$  into Eq. (6), we expect that the pulse advancement can be as large as the pulse duration. Such large advancement should be more than sufficient to test the predictions concerning the propagation of information described in Sec. I.

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